AI for Engineers

Understanding Before Using

Digital Engineers Research Team | March 2026

What Are We Actually Talking About?

A 60-second stack, bottom to top

The AI Stack

AI Agents	LLMs in a loop: read context, decide, use tools, check result, repeat
Large Language Models	Deep learning trained on text. A very sophisticated pattern completer
Deep Learning	ML with many layers of neurons. The 'deep' is literally 'more layers'
Machine Learning	Software that learns patterns from data instead of following hand-written rules

An agent is only as good as the model inside it, and the model is a pattern completer.

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Who here has used ChatGPT or Copilot? Who has used it for something beyond a simple question — generating code, writing a report, analyzing data?

AUDIENCE QUESTION

Part 1: The Problem

What happens when engineers use AI?

The Uncomfortable Numbers

84-90%

of software dev teams use AI daily

47%

experienced negative consequences

96%

don't fully trust AI, yet only 48% verify

Experience reveals problems rather than building confidence. The opposite of normal technology adoption.

Sources: Stack Overflow, JetBrains, McKinsey 2025, Sonar 2026

The Productivity Paradox

Context	Impact	Source
Lab (simple tasks)	**+55.8% faster**	Peng et al. 2023
Enterprise deployment	**+8.69%** **(+16% more build failures)**	Accenture 2024
Experienced devs, real codebases	**-19% slower**	METR 2025

19% slower with AI while perceiving themselves 20% faster.

A 39-percentage-point perception gap.

AI makes easy things easier.
But engineering is rarely easy.

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Show of hands — who uses AI tools in their daily work? And who has a systematic way of checking whether the output is correct?

AUDIENCE QUESTION

Beyond Software: "Real" Engineering

Chemical (HAZOP)

>86% textual similarity but only 19-37% semantically valid

Park et al. 2025

Civil (FE exam)

0% on Structural Design, 100% on Ethics

Naser et al. 2023

Mechanical

47-60% on core ME problems

Akolekar et al. 2025

Electrical (PCB)

AI hallucinated electronic components

JITX 2024

Pattern: AI produces outputs that look professionally competent but contain errors only domain experts catch.

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Has anyone here encountered AI output that looked correct at first glance but turned out to be wrong? What tipped you off?

AUDIENCE QUESTION

Part 2: Four Patterns

What's genuinely new when engineers work with AI agents?

Pattern 1: Learn the Material

LLMs have behavioral properties, like material properties in physical engineering.

Property	Example	Persistent?
Confident but wrong	6/22 claims over-confident	**Yes**
Can observe, can't score	Sees table, scores structure=1	**Yes**
Plausible-but-wrong output	Right units, right magnitude, wrong answer	Fading
Scores regress to mean	High work scored low, low scored high	Fading

You wouldn't use steel without knowing its fatigue properties. **Don't use LLMs without knowing theirs.**

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When you use a new material or tool for the first time — how do you learn its limits? Do you do the same with AI?

AUDIENCE QUESTION

Pattern 2: Layer Your Verification

V&V is old. V&V that costs minutes instead of months is new.

5	10	8	0
verification layers	defects found	missed by layer 1 alone	false positives

OPAL PCB Design Review

When verification is cheap, layer it. Run another pass. Audit the auditors.

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In your domain, how many independent checks does a design go through before it ships? What if you could add five more checks for the cost of a coffee?

AUDIENCE QUESTION

Pattern 3: Context Is Architecture

Agents have an auto-loading cliff: what's loaded gets used; everything else is invisible.

Above the cliff (auto-loaded)

Agent sees this, acts on it

Below the cliff

Invisible. Zero impact on behavior.

Doesn't work:

```
| API troubleshooting | docs/api-quirks.md |
```

Works:

```
| API returns 422 | docs/api-quirks.md |
```

Categories vs. situations. Agents recognize situations.

Pattern 4: Reproduce, Don't Assess

"Does this look right?" catches nothing. **"What answer do I get?" catches everything.**

Assessment

Claude + Gemini reviewed
68 equations

Errors found:

0

Reproduction

Same model, step-by-step

Errors found:

3

5x coupling error
84ms vs 43ms dwell time
12 s/hr stated as 36 s/hr

All three errors had correct units and plausible magnitudes — invisible to assessment.

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Think about how you review a colleague's calculations. Do you check the answer, or do you redo the calculation? What changes when the 'colleague' is an AI?

AUDIENCE QUESTION

Part 3: What This Means For You

Quick decision rules and honest limits

Quick Decision Rules

LLM gave me a number	Reproduce it	**Pattern 4**
LLM scored or rated something	Don't trust it — separate scoring	**Pattern 1**
LLM sounds very confident	Check the evidence tier	**Pattern 1**
Agent isn't using my docs	Check if it's below the cliff	**Pattern 3**
One review pass found nothing	Run another layer	**Pattern 2**

Honest Limits

- N=1 researcher, 9 projects, no controlled comparison
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- Two of four LLM properties may fade in 12-24 months
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- This is a practitioner's hypothesis, not a validated framework
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But: the underlying principles — verify, layer, design context, know your materials — are permanent. The specific techniques will evolve.

The hard part is no longer producing engineering work.
The hard part is knowing whether the work is any good.

Teaching engineers to use AI without teaching them to evaluate AI outputs is like teaching them to read test results without teaching them what the results mean.

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Starting tomorrow — which one of these four patterns could you apply to your current project?

AUDIENCE QUESTION

Thank You

Four Patterns for AI-Augmented Engineering

1. Learn the Material
2. Layer Your Verification
3. Context Is Architecture
4. Reproduce, Don't Assess